

Incident Case Report 001: LSASS Credential Dump and Cobalt Strike Injection

Case ID: INC-2026-001 **Analyst:** Detection Engineering **Date opened:** 2026-04-04 **Classification:** TRUE POSITIVE **Severity:** Critical

Executive Summary

Sysmon telemetry from the sampled corpus shows a confirmed LSASS credential dumping event executed remotely via WMI, followed weeks later by evidence of a Cobalt Strike beacon injected into `lsass.exe` on the same host family. The attacker had already achieved privileged access prior to the dump, given that the dumping process ran as `WmiPrvSE.exe` (SYSTEM context).

Timeline of Events (reconstructed from raw Sysmon evidence)

Time (UTC)	Source	Event	ATT&CK
2019-08-30 12:54:08.354	Sysmon EID 1	WmiPrvSE.exe spawns rundll32.exe with command: rundll32 C:\windows\system32\comsvcs.dll, MiniDump 4868 C:\Windows\System32\notepad.bin full	T1003.001, T1047
2019-08-30 12:54:08.x	Sysmon EID 11 (inferred)	Output file notepad.bin written to C:\Windows\System32\	T1003.001
2020-09-11 12:10:22.398	Sysmon EID 11	lsass.exe itself creates C:\Windows\System32\mimilsa.log	T1055.002, T1003.001

Root Cause Analysis

Initial vector

Not visible in this corpus slice, but the dumping process ran as `WmiPrvSE.exe`, meaning the attacker had **already obtained SYSTEM-level execution** on the host. Access was likely gained through lateral movement (see Hunt 001 Finding 2 — Impacket `wmiexec` signature).

Execution

The attacker abused the legitimate COM+ Services DLL `comsvcs.dll` which exposes a `MiniDump` export. Calling it via `rundll32` produces a full memory dump of the target process without dropping a dedicated tool binary.

- Target PID 4868 = LSASS
- Output `notepad.bin` = public PoC default; chosen for innocuous filename
- `full` argument = complete memory dump including cached credential material

Post-exploitation

The `mimilsa.log` file created by `lsass.exe` on 2020-09-11 is the **default log filename emitted by mimikatz when run inside a Cobalt Strike beacon** (`beacon> mimikatz / beacon> log mimilsa`). The file being written by **`lsass.exe`** rather than by a separate process proves the beacon was injected into LSASS memory, not just running alongside it.

Evidence

Primary event (the dump)

```
```json { "time": "2019-08-30T12:54:08.354049Z", "eid": 1, "provider":  
"Microsoft-Windows-Sysmon", "image": "C:
\\Windows\\System32\\rundll32.exe", "cmd": "rundll32 C:
\\windows\\system32\\comsvcs.dll, MiniDump 4868 C:
\\Windows\\System32\\notepad.bin full", "parent": "C:
```

\\Windows\\System32\\wbem\\WmiPrvSE.exe" } EOF cat > incidents/  
incident\_001\_lsass\_dump\_and\_cobalt\_strike.md <<'MDEOF' # Incident  
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\\Windows\\System32\\wbem\\WmiPrvSE.exe" } { "time":
```

“2020-09-11T12:10:22.398726Z”, “eid”: 11, “provider”: “Microsoft-Windows-Sysmon”, “image”: “C:\Windows\system32\lsass.exe”, “file”: “C:\Windows\System32\mimilsa.log” } Detection rule that fired

win\_comsvcs\_minidump.yml (ID eeeeeeee-5555-5555-5555-555555555555)  
– matches the primary event.

win\_lsass\_mimikatz\_log\_artifact.yml (ID  
10000007-0007-0007-0007-000000000007) – matches the CS artifact.

Impact Assessment Asset Impact Host Complete compromise (SYSTEM)  
Credentials LSASS cached domain credentials for the host’s logged-on  
users Domain Potential lateral movement using harvested credentials  
Persistence Cobalt Strike beacon resident in LSASS memory; re-infects  
on reboot if a persistent mechanism exists Closure Classification

TRUE POSITIVE, justified by:

Corroborating evidence from two Sysmon event types (EID 1 and EID  
11) across two timelines.

Presence of a known public PoC command line (comsvcs MiniDump)  
with no legitimate business justification.

Presence of a mimikatz/Cobalt Strike artifact file created by  
LSASS – no benign explanation.

### Detection Gaps Revealed

No detection for rundll32 being spawned by WmiPrvSE.exe. Rule  
win\_wmi\_process\_call\_create.yml catches any WmiPrvSE child but  
with a medium severity; the combination with rundll32 and comsvcs  
should be its own high-severity rule (covered by  
win\_comsvcs\_minidump.yml).

No detection for LSASS writing unusual files. Rule  
win\_lsass\_mimikatz\_log\_artifact.yml closes this specific gap but  
only for the two filenames observed. Broader "LSASS writes any  
file to System32" rule could follow.

Initial access vector not detected in this corpus slice. Further  
hunt required.

### Recommended Response

See playbooks/credential\_dumping.md. Immediate actions:

Isolate host from network.

Force password reset for all accounts with cached credentials on the host.

Rotate KRBTGT if domain admin credentials were cached.

Search fleet-wide for mimilsa.log and other mimikatz artifact filenames.

Reimage host before returning to production.

### ATT&CK Mapping

T1003.001 – OS Credential Dumping: LSASS Memory

T1047 – Windows Management Instrumentation

T1055.002 – Process Injection: Portable Executable Injection

T1036.005 – Masquerading: Match Legitimate Name or Location  
(notepad.bin path)

### Evidence Files

incidents/incident\_001\_evidence/comsvcs\_events.jsonl – 50 events,  
all LSASS-related Sysmon records from the corpus

hunts/hunt\_001\_evidence/all\_proc\_creation.jsonl – parent process  
chain

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